

Tutorial-08

CID	x_1 Credit Score	x_2 Annual Income	x_3 Emp Years	x_4 loan amount	Approved
1	750	80	8	200	A
2	680	35	2	150	R
3	780	95	10	250	A
4	580	30	1	180	R
5	720	75	7	180	A
6	620	40	3	160	R

	y_1	y_2	y_3	y_4	
T_1	700	70	6	190	A
T_2	610	38	2	155	R
T_3	770	90	9	210	A

$$\begin{aligned}
 D_{11} &= \sqrt{(x_1 - y_1)^2 + (x_2 - y_2)^2 + (x_3 - y_3)^2 + (x_4 - y_4)^2} \\
 &= \sqrt{(750 - 700)^2 + (80 - 70)^2 + (8 - 6)^2 + (200 - 190)^2} \\
 &= 52
 \end{aligned}$$

$$2 = \sqrt{(680 - 700)^2 + (35 - 70)^2 + (2 - 6)^2 + (150 - 190)^2} = 113.3$$

$$3 = \sqrt{(780 - 700)^2 + (95 - 70)^2 + (10 - 6)^2 + (250 - 190)^2} = 103.75$$

$$4 = \sqrt{26.98}$$

$$5 = 22.9$$

$$6 = 90.60$$

Accuracy

Next Step do with T_2 & y values are T_2

$$1 = \sqrt{(780-610)^2 + (80-38)^2 + (8-2)^2 + (200-153)^2} = 153.2$$

$$2 = \sqrt{(680-610)^2 + (35-38)^2 + (2-2)^2 + (150-153)^2} = 11.37$$

$$3 = \sqrt{(780-610)^2 + (95-38)^2 + (10-2)^2 + (250-153)^2} = 203.87$$

$$4 = \sqrt{(580-610)^2 + (30-38)^2 + (1-2)^2 + (180-153)^2} = 39.12$$

$$5 = \sqrt{(780-610)^2 + (75-38)^2 + (7-2)^2 + (180-153)^2} = 118$$

$$6 = \sqrt{(620-610)^2 + (40-38)^2 + (2-3)^2 + (155-160)^2} = 11.40$$

Next Step do with T_3 & y values are T_3

$$1 = \sqrt{(750-770)^2 + (80-90)^2 + (8-9)^2 + (200-230)^2} = 37.42$$

$$2 = \sqrt{(680-770)^2 + (35-90)^2 + (2-9)^2 + (150-230)^2} = 195.29$$

$$3 = \sqrt{(780-770)^2 + (95-90)^2 + (10-9)^2 + (250-230)^2} = 22.93$$

$$4 = \sqrt{(580-770)^2 + (30-90)^2 + (1-9)^2 + (180-230)^2} = 105.57$$

$$5 = \sqrt{(780-770)^2 + (75-90)^2 + (7-9)^2 + (180-230)^2} = 72.31$$

$$6 = \sqrt{(620-770)^2 + (40-90)^2 + (3-9)^2 + (160-230)^2} = 143.02$$